

Chronic Photo-Oxidative Stress and Subsequent Macrophage Chemo-attractant Protein-1 Activation as Causative Factors for Age-Related Macular Degeneration

An analysis of the research authored by Suzuki et. al., and suggested follow up experiments to continue the research of the team.

I: Goals and Hypotheses

Age-related macular degeneration (AMD) and other senescence related diseases are conditions that commonly plague our elderly population. In many cases, multiple different factors contribute to the pathogenesis of these conditions. In this research article, the photo-oxidation of membrane lipids coupled with the activation of monocyte chemo-attractant proteins is explored as a possible model for the onset of this disease. The authors used prolonged photic stress to illicit phospholipid oxidation and subsequent inflammatory processes that mimicked Human AMD in mice. The authors then used consecutive experiments to establish a pathogenic pathway involving these factors that culminated in the onset of AMD. Finally, the authors analyze the data and suggest protection from photo oxidative stress and suppression of inflammation as new therapeutic targets for age-related macular degeneration.

II: Models of Experimentation

In order to establish the pathogenic pathway of AMD, the authors had to use several different models including gene knockout mice, gene knockdown mice, low intensity blue light irradiated mice, oxidized phospholipid injected mice, and different combinations of the above. For example, gene knockout mice were irradiated in some experiments to test the effects of the presence of gene products in combination with photic stress on the onset of macular degeneration, etc. The eyes of each of these mouse models were isolated and then the retinas were cryo-sectioned for further analysis (usually immunohistochemistry or global staining). Details of these different mouse models are specified below.

1. To prove that long term exposure to normal levels of light were sufficient to induce photo oxidation of membrane lipids, wild type mice were exposed to long term low intensity photic stress [stuck in a cage that was bathed in blue LED light continuously]. This model of inducing photic stress is meant to mimic the effects of human exposure to artificial light during most of our waking hours. This model is what is referred to throughout the paper when long term low intensity irradiation is mentioned. Gene knockout mice, gene knockdown mice, were both exposed to this long term irradiation to see the different

combinatory effects of the lack of gene products and photic stress. Spectrometric analysis of membrane lipids from wild-type mice revealed that this irradiation was sufficient to cause photo oxidative stress as was revealed by the presence of various esters of phosphatidylcholine. Note that photo oxidative stress induced the oxidation of membrane lipids from all layers of the retina, not just the choroid or the retinal pigment epithelium.

2. To prove that long term low-light irradiation was sufficient to induce an increase of monocyte chemoattractant protein-1 (abbreviated MCP-1), a protein that essentially serves as a chemoattractant for macrophages, mouse retinas from irradiated mice and wild type were immunostained for MCP-1.
3. To prove that photo-oxidized membrane lipids were sufficient to cause choroidal neovascularization (one of the final stages of AMD), oxidized membrane lipids were injected into the sub retinal space of wild-type mice and MCP-1 and MCP-2 knockout mice. These results show that photo oxidized membrane lipids are sufficient to cause AMD when injected into the retina of wild type mice. These results also show that MCP proteins are necessary in the pathogenesis of AMD, because in MCP knockout mice, the injection of photo oxidized membrane lipids was not sufficient to cause the symptoms of AMD.
4. To prove that long-term, low intensity photo oxidative stress induces basal laminar deposits coupled with macrophage accumulation and choroidal neovascularization (The hallmark symptoms of human AMD), both wild type mice, MCP-1^{-/-}, and CCR-2^{-/-} mice were exposed to the low level irradiation levels of part 1. Indeed, wild type mice exhibited the symptoms of human AMD while MCP-1 and CCR-2 knockout mice did not exhibit choroidal neovascularization [CNV].
5. To unambiguously prove that the oxidation of membrane lipids was the cause of the choroidal neovascularization, α -phenyl-N-tert-butyl nitron (PBN), an antioxidant, was daily injected into wild type mice during light irradiation. The resulting retinas were immunostained against MCP-1 proteins and it was found that treatment of irradiated mice with PBN significantly reduced the expression of MCP-1 and halted the higher level symptoms of AMD.
6. To see if MCP-1 activation induces the expression of VEGF and PEDF, (two other proteins which have been implicated in previous studies to be present in AMD patients) immuno-histochemistry was performed on irradiated mouse retinas with anti-VEGF and anti- PEDF antibodies. VEGF is an angiogenic growth factor (implicated in the growth of new blood vessels) and PEDF is thought to be an anti-angiogenic growth factor (blocks the formation of new blood vessels). Photooxidation led to VEGF overexpression and PEDF under expression. In addition, treatment of irradiated cells with MCP-1 siRNA significantly decreased the expression of VEGF which suggests that VEGF is further down the pathway of pathogenesis of AMD than MCP-1.

III. Methods of Experimentation:

Spectrometric Analysis of Phospholipids

Lipid extracts from the retina were isolated and dissolved in a methanol/water solution (90:10). These membrane lipids were then subjected to chromatography– electrospray ionization tandem mass spectrometry. From my understanding, they used really high voltage electric currents to ionize the solvent system (membrane lipids suspended in the methanol-water system) and then converted it into an aerosol form in which the authors then performed a spectrometric analysis on. This technique was what allowed the authors to establish that long term low intensity photic stress induced membrane lipids in the choroid to become photo-oxidized.

Immunohistochemistry

Antibody staining of the tissues revealed if certain gene products were present in the tissue at the time of the staining. The benefits of such an immunohistochemistry experiment determine the presence of a gene product in vivo. The Independent variable in such an experiment is whether or not the gene product is present in the tissue. The dependent variable is whether or not you witness the glowing of the fluorescent antibodies, for the gene product needs to be present in the tissue for the antibodies to bind and subsequently, for the experimenters to witness a fluorescence.

To determine if the photo oxidation of membrane lipids caused the expression of MCP-1, tissues isolated from the irradiated mice were treated with anti MCP-1 immunofluorescent antibodies. The tissue sections were then examined, and results were drawn. This method was used to see if chronic photo oxidation induced MCP-1. Indeed immunohistochemistry revealed the presence of MCP-1 in the retinas of irradiated mice but not wild type mice. This is consistent with the authors' hypotheses.

This technique was also used to see if the presence of photo oxidized lipids were sufficient to induce MCP-1 activation. During this part of the experiment, none of the mice underwent the constant low level irradiation. Instead, photo oxidized membrane lipids were injected into the sub retinal space of the mice, and immunohistochemistry was used to detect the presence of MCP-1 within these tissues, by comparing the presence of the antibody staining of the injected mice tissues with that of just the normal wild type mice. What the authors found is that the injection of the photooxidized membrane lipids without the irradiation itself was sufficient to cause MCP-1 expression and subsequently the pathogenesis of AMD. This part of the experiment ruled out that the irradiation of the mice retinas caused Macular degeneration by some other sort of pathogenic pathway.

It is possible that photo oxidation can cause the effects of AMD by the activation of many different pathogenic pathways in parallel, not just the MCP-1 pathway. To prove that the photo oxidation of membrane lipids caused the symptoms of macular degeneration by the MCP-1 pathway alone, irradiated mice were injected with α -phenyl-N-tert-butyl nitrone (PBN), an antioxidant, daily during light irradiation. The retinas of these mice were then immunostained against MCP-1 proteins and it was found that treatment of irradiated mice with PBN significantly reduced the expression of MCP-1 and halted the higher level symptoms of AMD.

The growth factors VEGF and PEDF have been found to be present in AMD afflicted retinas in the past. To test whether VEGF and PEDF were being expressed as a result of the photo oxidative stress induced in the mice retinas of this experiment; immunohistochemistry was used with polyclonal antibodies against these two secretory proteins. The authors found that the photo oxidation of the mice led to the expression of VEGF and the under expression of PEDF. These retinas were of course tested against wild type mice that had not been irradiated. To see if the expression of MCP-1 was causal in the expression of these two proteins, the same antibody stains against VEGF and PEDF were used in knock down experiments of MCP-1 to see if a decrease in MCP-1 would in turn cause a change in the expression of these proteins. What the authors found was that the use of siRNA to knock down the expression of MCP-1 caused a decrease in the expression of VEGF. The fact both of these proteins are implicated in the pathogenesis of the AMD but that the knock down of MCP-1 causes a drop in the expression of VEGF indicates that VEGF is down stream of MCP-1 in the pathogenesis of AMD.

Knock out Mice

The authors induced MCP-1 knock out mice to prove that the protein in question was important in the pathogenesis of macular degeneration. A knock out mouse is an animal where the gene for a certain protein, in this case MCP-1, is taken out of that mouse's genome or rendered nonfunctional. The benefits of this are that an experimenter can see the changes that a certain gene product elicits. The independent variable of this type of experiment is the presence or absence of a gene as determined by the experimenters. The dependent variable is the effect that this knockout has on global tissue structure, chemical cascades, pathogenic pathways etc.

In this particular gene knock out experiment, the authors test if irradiation of the MCP-1 knock out mice exhibited the same AMD characteristics as the normally irradiated mice. If the MCP-1 knockout mouse exhibited the same characteristics, then we know that the pathogenesis of the AMD disease follows a different pathogenic pathway than the one the authors thought. If instead, the knocking out of the MCP-1 gene causes a halt in the pathogenesis of the AMD condition despite the irradiation that the mice receive, then we know that MCP-1 is indeed an important part of the pathogenesis of this disease.

Gene Knock Down Experiment

In addition to the gene knock out experiment, the authors of this paper did a gene knock down experiment in which they tested the effect of MCP-1 on the expression of the growth factors VEGF and PEDF. A gene knock down experiment is one in which short strands of complimentary sequences to the mRNA of a gene product are introduced into the cell. The small interfering RNA then binds to the mRNA in question and as a result, the mRNA which would usually be translated into protein is mistaken as foreign and degraded in a process catalyzed by the proteins dicer and argonaut. The independent variable in this type of experiment is again the presence or absence of a gene product. The dependent variable is the effect of this presence/absence on the chemical cascades, pathogenesis, etc. of a particular mouse.

The authors of the paper used siRNA to stop the translation of MCP-1 mRNA into protein, thus testing if the VEGF and PEDF which in previous experiments were proved to be part of the same pathogenic pathway were either upstream or downstream of the expression of MCP-1. The benefits of this experiment are that it is cheaper and faster to perform than a gene knock out experiment. Some of the disadvantages to this technique are that it does not guarantee that all of the mRNA in question gets degraded. In this case, it would greatly decrease the amount of MCP-1 in the retina, but it would not abolish the presence of the protein completely. So for studies where an experimenter is studying the absence of a particular protein, a gene knock down experiment would not be ideal. However, for the purposes of this experiment, the RNAi is perfect because the authors were just trying to determine if the decrease in the number of MCP-1 protein transcripts would have an effect on the downstream pathogenesis, not abolish it completely.

Histological Examination of Retinal Tissue

Microscopic examination of tissues allows experimenters to see the histological effects of knockout experiments and other procedures done at the genetic level. The independent variable would be the gene knockouts/conditions/whatever the experimenters really wanted to study on the tissue in question. The dependent variable would in turn be the macroscopic structure of the tissue being studied.

In this particular experiment, histological examination of the retinal tissue at advanced stages of macular degeneration reveals choroidal neovascularization in the layers below the retinal pigment epithelium. Choroidal neovascularization, or the recruitment of blood vessels into the choroid (the layer of connective tissue located below the retinal pigmented epithelium) is a hallmark of late stage macular degeneration, and is easily picked up by a simple Hematoxylin and Eosin [H&E] stain. The simplicity of this experiment coupled with its ability to easily visualize structural changes in a tissue makes it a powerful addition to the cavalry of other experiments at the authors' disposal.

IV. Results

During the course of their experiment, the authors proved that long term exposure to low intensity light induces photo oxidation in the retina. This was proved when wild type mice were subjected to irradiation; membrane lipids were isolated from those mice and subjected to the chromatography–electrospray ionization tandem mass spectrometry which revealed oxidized ester analogues of phosphatidylcholine.

The authors then proved that the oxidized phospholipids brought on by the light irradiation in vivo were sufficient to induce the expression of MCP-1. This was proved by the immunohistochemistry of tissue sections isolated from irradiated mice.

Next, the authors proved that the presence of oxidized membrane lipids, and not the light irradiation itself was the cause of the symptoms of AMD. This was proved by the experiment in which the authors injected photooxidized lipids into the sub retinal space of non irradiated mice. The injection caused the expression of MCP-1 and other chemo attractant proteins, subsequently inducing the onset of AMD symptoms without the exposure to the long term low level light irradiation. This was observed in immunohistochemistry and histological

examinations of the retinal tissues.

The authors proved that the pathogenesis of AMD was dependent on the expression of MCP-1 and the changes that it produced instead of some other process that could have been brought on in parallel by the very same exposure to light. The authors proved all of this by producing MCP-1 $-/-$ knock out mice and Ccr2 $-/-$ knockout mice. They then subjected these mice to long term low level irradiation and compared the histological changes (tested by the H&E stain) of these knockout mice to that of wild type mice that underwent the same process. What they found was that while the wild type mice were observed to have choroidal neovascularization and other hallmark symptoms of AMD, the knockout mice did not exhibit any of these phenotypes when exposed to the same irradiation. This proves that while long term photo oxidation of membrane lipids caused the symptoms of AMD, and that these symptoms were dependent on the expression of MCP-1 and Ccl2 proteins.

The authors suggested antioxidants as effective therapeutic agents for the reversal of AMD. To prove this, the authors gave mice daily injections of α -phenyl-N-tert-butyl nitrone (PBN), an antioxidant, during light irradiation. This treatment significantly reduced the expression of MCP-1 and halted the higher level symptoms of AMD brought on further downstream the pathogenic pathway.

Finally, the authors proved that MCP-1 activation induces the expression of VEGF, an angiogenic growth factor and reduces the expression of PEDF, an anti-angiogenic growth factor. Immunohistochemistry was used to reveal that VEGF was expressed in irradiated mouse retinae while PEDF was under expressed. Furthermore, it was proved that VEGF is a protein that is expressed downstream of MCP-1's effects by injecting irradiated mice with MCP-1 siRNA, and then doing immunostaining for VEGF. The results of this particular experiment were that VEGF was significantly decreased when the irradiated cells were treated with the MCP-1 knockdown. These experiments proved the authors' hypotheses correct and even revealed therapeutic targets for the treatment of AMD.

V: Thoughts and Comments

Overall, the authors did an exceptionally thorough experiment in which they unambiguously implicated the photo oxidation of membrane lipids in the pathogenesis of age related macular degeneration. They proved that the oxidation of these phospholipids, which was brought on by photic stress, caused the expression of MCP-1 and Ccl2, which in turn serve as chemo attractants for macrophages. They identified VEGF as another protein that was implicated in the pathogenesis of AMD, and identified VEGF's position in the pathogenic pathway of AMD as being downstream of MCP-1. They even identified possible therapeutic targets for the treatment of this disease. Although the authors implicated these proteins in the pathogenesis of AMD they failed to elucidate the *entire* pathogenic pathway for the onset of AMD. For example, *how* does the photo oxidation of membrane lipids increase the expression of MCP-1? How is this signaled to the nucleus and what transcription factors are effected? How do these soluble proteins like MCP-1 and VEGF illicit the changes that take place for AMD by recruiting macrophages and causing neovascularization? Although this study sheds significant light on the factors governing the onset of macular degeneration, and the authors' experiments were appropriate in answering

their questions, much more experimentation is needed.

The next part of the pathway that I would study would be how MCP-1 recruits macrophages to cause the next step of macular degeneration. In this I research possible receptors for MCP-1 on macrophages. Then I would do a Co-Immunoprecipitation experiment whereby I would expose the different receptors to MCP-1 and then after washing off the excess chemo attractant, I would treat the receptors with anti-MCP-1 fluorescent antibodies. If the MCP-1 bound to the receptors in question, then they would fluoresce when treated with the anti-MCP-1 antibody. If MCP-1 did not bind to the receptors; however, there would be no fluorescence because the MCP-1 would be washed away if it did not bind to the receptors.

References:

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